

MATH 4720 / MSSC 5720

Instructor: Mehdi Maadooliat

Lecture 1



Department of Mathematical and Statistical Sciences



SYLLABUS – WEBSITE - D2L

- Syllabus
- Course Materials in <http://tinyurl.com/f26-4720>
- Homework and Discussions: <http://d2l.mu.edu>



Q

Course information ▾

- Overview
- Syllabus
- Schedule
- Useful links

HW >

Computing >

Course information > Schedule

Statistical Methods

This page contains an outline of the topics, content, and assignments for the semester. Note that this schedule will be updated as the semester progresses and the timeline of topics and assignments might be updated throughout the semester.



D2L: Marquette University's Learning Management System

Use your CheckMarq username and password to log in. Trouble logging in? You can [reset your password](#) or contact the [IT Services Help Desk](#).

Username *

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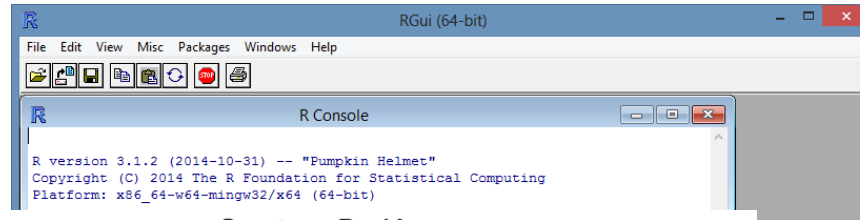
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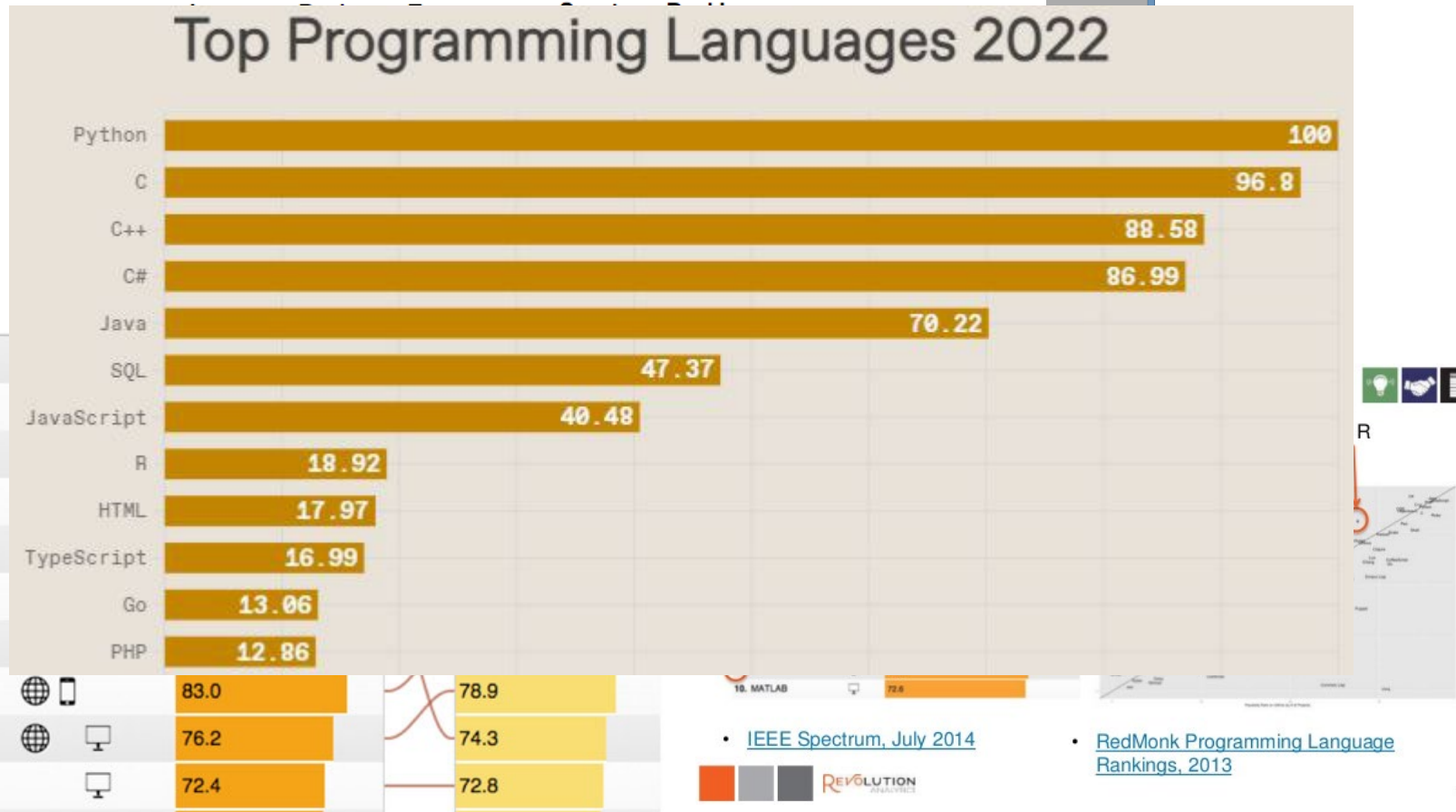
R IS THE STATISTICAL SOFTWARE THAT WE MOSTLY USE IN THIS COURSE



- It is free!!



[IEEE Spectrum, July 2016](#)



[IEEE Spectrum, July 2015](#)



HISTORY OF R (FROM [WIKIPEDIA](#))

- R is a [programming language](#) and software environment for [statistical computing](#) and graphics. The R language is widely used among [statisticians](#) and [data miners](#) for developing [statistical software](#) and data analysis. Polls, [surveys of data miners](#), and studies of scholarly literature databases show that R's popularity has increased substantially in recent years.
- R is an implementation of the [S programming language](#) combined with [lexical scoping](#) semantics inspired by [Scheme](#). [S](#) was created by [John Chambers](#) while at [Bell Labs](#). There are some important differences, but much of the code written for S runs unaltered.
- R was created by [Ross Ihaka](#) and [Robert Gentleman](#) at the [University of Auckland](#), New Zealand, and is currently developed by the *R Development Core Team*, of which Chambers is a member. R is named partly after the first names of the first two R authors and partly as a play on the name of [S](#).
- R is a [GNU project](#). The [source code](#) for the R software environment is written primarily in [C](#), [Fortran](#), and R. R is freely available under the [GNU General Public License](#), and pre-compiled binary versions are provided for various [operating systems](#). R uses a [command line interface](#); there are also several [graphical front-ends](#) for it.

R: THE STATISTICAL SOFTWARE



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Be The Difference.

- Download and Install:

Windows

Comprehensive R Archive Network

Mac

The screenshot displays the RStudio environment with several windows open:

- R Console:** Shows the execution of R code, including data loading and plotting commands.
- R Editor:** Displays the R script being edited, with line numbers and syntax highlighting.
- R Workspace Browser:** Lists the objects in the current workspace, including their types and dimensions.
- R Package Manager:** Shows the installed and available packages, including the R Graphics Package.

The R Console output includes the following code and results:

```
> team2003$teamID
[1] ANA ARI ATL BAL BOS CHA CHN CIN CLE COL DET FLO HOU KCA LAN MIL MIN MON NYA
[20] NYN OAK PHI PIT SIN SEA SFN SLN TBA TEX TOR
148 Levels: ALT ANA ARI ATL BAL BFN BFP BL1 BL2 BL3 BL4 BLA BLF BLN BLU ... WSO
> team2003$payroll
[1] 79031667 80657000 106243667 73877500 99946500 51010000 79060313
[8] 59355667 48504834 67179667 49168000 49450000 71040000 40510000
[15] 105572620 40627000 55505000 51948500 152749014 116876429 50260813
[22] 70780000 54812429 45210000 86959167 82852167 83786666 19630000
[29] 103491667 51269000
> fitted(team2003$salary,lm)
1 2 3 4 5 6 7 8
82.58287 82.90759 86.01950 81.55313 86.76340 76.98448 82.75003 76.65184
9 10 11 12 13 14 15 16
76.49996 80.21498 76.61647 76.67281 80.98623 74.08803 87.88544 74.91008
17 18 19 20 21 22 23 24
77.08252 77.17198 97.31087 90.14380 76.83480 80.93429 77.74416 75.82571
25 26 27 28 29 30
84.16669 83.34616 83.53286 70.71513 87.46969 77.03622
> team2003$W
[1] 77 84 101 71 95 86 88 69 68 74 43 91 87 83 85 68 90
[20] 66 96 86 75 64 93 100 85 63 71 86
> pairs(team2003$forpairs)
> edit(team2003)
> formatPlot.R
1 library(ggplot2)
2 source("plots/formatPlot.R")
3
4 view(diamonds)
5 summary(diamonds)
6
7 summary(diamonds$price)
8 aveSize <- round(mean(diamonds$carat), 4)
9 clarity <- levels(diamonds$clarity)
10
11 p <- qplot(carat, price,
12 data=diamonds, color=clarity,
13 xlab="Carat", ylab="Price",
14 main="Diamond Pricing")
15
16 format.plot(p, size=24)
```

The R Console output shows the summary of the diamonds dataset:

```
summary(diamonds$price)
Min. 1st Qu. Median Mean 3rd Qu. Max.
326 950 2401 3933 5324 18820
> aveSize <- round(mean(diamonds$carat), 4)
> clarity <- levels(diamonds$clarity)
> p <- qplot(carat, price,
+ data=diamonds, color=clarity,
+ xlab="Carat", ylab="Price",
+ main="Diamond Pricing")
> format.plot(p, size=24)
```

The R Editor shows the following code:

```
1 library(ggplot2)
2 source("plots/formatPlot.R")
3
4 view(diamonds)
5 summary(diamonds)
6
7 summary(diamonds$price)
8 aveSize <- round(mean(diamonds$carat), 4)
9 clarity <- levels(diamonds$clarity)
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11 p <- qplot(carat, price,
12 data=diamonds, color=clarity,
13 xlab="Carat", ylab="Price",
14 main="Diamond Pricing")
15
16 format.plot(p, size=24)
```

The R Workspace Browser shows the following objects:

Object	Type	dim:
data	data.frame	dim: 20 4
g	factor	levels: 10
i	numeric	length: 12
n	numeric	length: 1
opar	list	length: 2
pie.sales	numeric	length: 6
pin	numeric	length: 2
scale	numeric	length: 1
usr	numeric	length: 4
women	data.frame	dim: 15 2
height	numeric	length: 15
weight	numeric	length: 15
x	numeric	length: 87

The R Package Manager shows the following packages:

status	Package	Description
loaded	graphics	The R Graphics Package
not loaded	grid	The Grid Graphics Package
not loaded	lattice	Lattice Graphics
loaded	methods	Formal Methods and Classes
not loaded	monoc	CAME with C/C++ interfaces, animation

The R Package Manager also shows the R Graphics Package version 2.0.0 and the help pages.

- RStudio:

- [Download](#)



MATH 4720/MSSC 5720 CONT...

- **Any General questions about Homework, and Projects:**

- **SHOULD** be posted in D2L Discussion Board.
- I will **NOT** answer general emails about Homework and/or Projects.

- **Homework and Projects:**

- Should be submitted as a **PDF** file (**Otherwise you will get ZERO**):
 - How to Combine Images into a PDF file [FREE & EASY + No Software] ([Youtube](#))
 - Microsoft Word to PDF in 10 Seconds ([Youtube](#))
 - How to: convert Images to PDF in Macbook/iMac ([Youtube](#))
 - <http://apple.stackexchange.com/questions/11163/how-do-i-combine-two-or-more-images-to-get-a-single-pdf-file>

MATH 1700 102 Mod Elementary Stat		
Course Home	Content	Discussions
Dropbox	Quizzes	Classlist
Grades		
Discussions List	Subscriptions	Group and Section Restrictions
Statistics		
New	More Actions	
Filter by:	Unread	Unapproved
Homeworks		
Hide Topics for Homeworks		
Topic	Threads	Posts
Homework 1	0	0
Homework 2	0	0
Homework 3	0	0
Homework 4	0	0
Homework 5	0	0
Homework 6	0	0
Homework 7	0	0
Homework 8	0	0



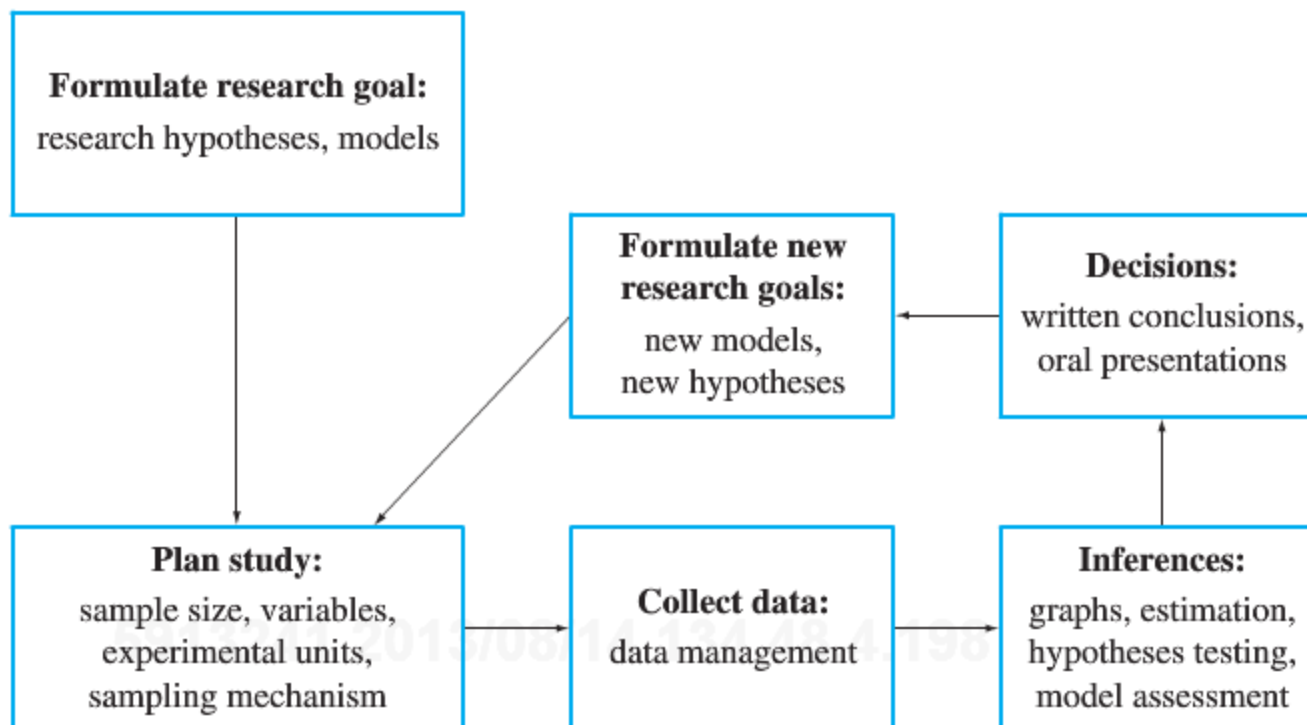
TOPIC 1: CHAPTER 1-2

- **Statistics and Data Description**
- **Populations and Samples**
- **Types of Studies**
- **Confounding Variable**

WHAT IS STATISTICS?

- What do you think of when you hear the word “statistics”?
- **Statistics:** The science of collecting, classifying, and interpreting data.
- Anticipated learning outcomes:
 - appreciate and apply basic statistical methods **in their scientific field**
 - appreciate and apply basic statistical methods **in an everyday life setting**

HOW TO LEARN FROM DATA?

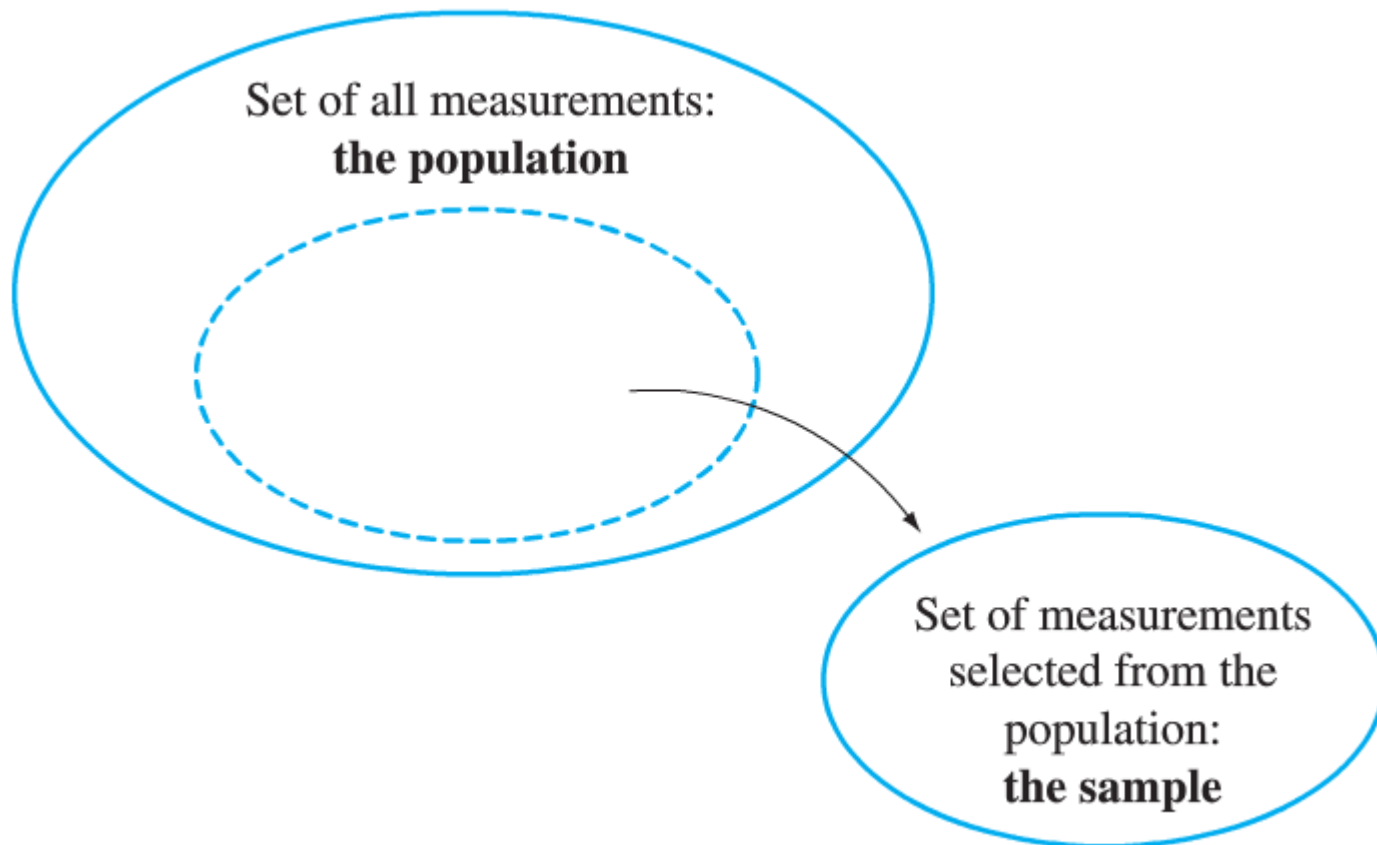


WHAT SHOULD YOU EXPECT?

The Four-Step Process	Chapters
1 Defining the Problem	1 Statistics and the Scientific Method
2 Collecting the Data	2 Using Surveys and Experimental Studies to Gather Data
3 Summarizing the Data	3 Data Description
	4 Probability and Probability Distributions
4 Analyzing the Data, Interpreting the Analyses, and Communicating the Results	5 Inferences about Population Central Values
	6 Inferences Comparing Two Population Central Values
	7 Inferences about Population Variances
	8 Inferences about More Than Two Population Central Values
	9 Multiple Comparisons
	10 Categorical Data
	11 Linear Regression and Correlation
	12 Multiple Regression and the General Linear Model

POPULATION VS. SAMPLE

- **Population:** The entire group of interest
- **Sample:** A part of the population selected to draw conclusions about the entire population



COLLECTING DATA

- **Observational study:** Observe a group and measure quantities of interest. This is passive data collection in that one does not attempt to influence the group. The purpose of the study is to describe the group.
- **Experiment:** Deliberately impose treatments on groups in order to observe responses. The purpose is to study whether the treatments cause a change in the responses.

EXPERIMENT TERMS

- **Experimental Group:** A collection of experimental units subjected to a real treatment.
- **Control Group:** A collection of experimental units subjected to the same conditions as those in an experimental group except that no treatment is imposed.
- This design helps control for potential **confounding** effects.

WHAT IS **CONFOUNDING**?

- The Variables that are not in control of the researcher.
- A variable that is not among the **explanatory** or **response** variables in a study and yet may influence the interpretation of relationships among those variables.
- A perceived relationship between an **dependent(response)** variable and a **independent(explanatory)** variable that has been misestimated due to the failure to account for a **confounding factor** is termed a spurious relationship
 - Socioeconomic status and Life expectancy
 - Berkeley gender bias case (http://en.wikipedia.org/wiki/Simpson's_Paradox)

LURKING VARIABLE AND SIMPSON'S PARADOX

- **Lurking variable:** A variable that is not included in a study but has an effect on the variables of the study and makes it appear that those variables are related.
- **Simpson's Paradox:** An association or comparison that holds for all of several groups can **reverse direction** when a **lurking variable** is present.
- **Example: Kidney stone treatment**(Br Med J (Clln Res Ed) 292 (6524): 879-882)

	Treatment A	Treatment B
Small Stones	<i>Group 1</i> 93% (81/87)	<i>Group 2</i> 87% (234/270)
Large Stones	<i>Group 3</i> 73% (192/263)	<i>Group 4</i> 69% (55/80)
Both	78% (273/350)	83% (289/350)

- http://en.wikipedia.org/wiki/Simpson's_Paradox



ASSOCIATION BETWEEN CELL PHONE USE AND THE OCCURRENCE OF CANCER?

- **Recent USA Today article.**
- **Three studies:**
 - **A German study (Stang et al., 2001) compared 118 patients with a rare form of eye cancer to 475 healthy patients who did not have the eye cancer. The patients cell phone use was measured using a questionnaire. The eye cancer patients used cell phones more often, on the average.**
 - **A British study (Hepworth et al., 2006) compared 966 patients with brain cancer to 1716 patients who did not have brain cancer. The patients cell phone use was measured using a questionnaire. The two groups' use of cell phones was similar.**
 - **An Australian study (Repacholi, 1997) conducted an experiment with 200 transgenic mice, specially bred to be susceptible to cancers of the immune system. One hundred mice were exposed for two-half hour periods a day to the same kind of microwaves with roughly the same power as that transmitted from a cell phone. The other 100 were not exposed. After 18 months, the brain tumor rate for the mice exposed to radiation was twice as high as the brain tumor rate for the unexposed mice.**

REFERENCES FOR CELL PHONE & CANCER STUDIES

- **Hepworth, SJ et al., (2006) Mobile phone use and risk of glioma in adults: case control study. British Medical Journal, 332, 883-887.**
- **Repacholi, HM (1997) Radio frequency field exposure and cancer. Environ. Health Prospect, 105, 1565-1568.**
- **Stang A et al., (2001) The possible role of radio frequency radiation in the development of uveal melanoma. Epidemiology, 12(1), 7-12.**

QUESTIONS TO CONSIDER ABOUT THESE THREE STUDIES

- **How do the three studies differ?**
 - In studies 1 and 2 no treatments are assigned. Patients are merely questioned. Thus, studies 1 and 2 are observational studies.
 - Study 3 uses experiments on mice with the hope of generalizing to humans.
- **Why do the results of different medical studies sometimes disagree?**
 - Differing types of studies, data collection, sample frames.
 - Sampling variability.
- **Could the third study have used human subjects instead?**
 - No, because it would be unethical to knowingly expose humans to possibly harmful waves.

INFERENCE STATISTICS

- **Example (1988, the Steering Committee of the Physicians' Health Study Research Group)**

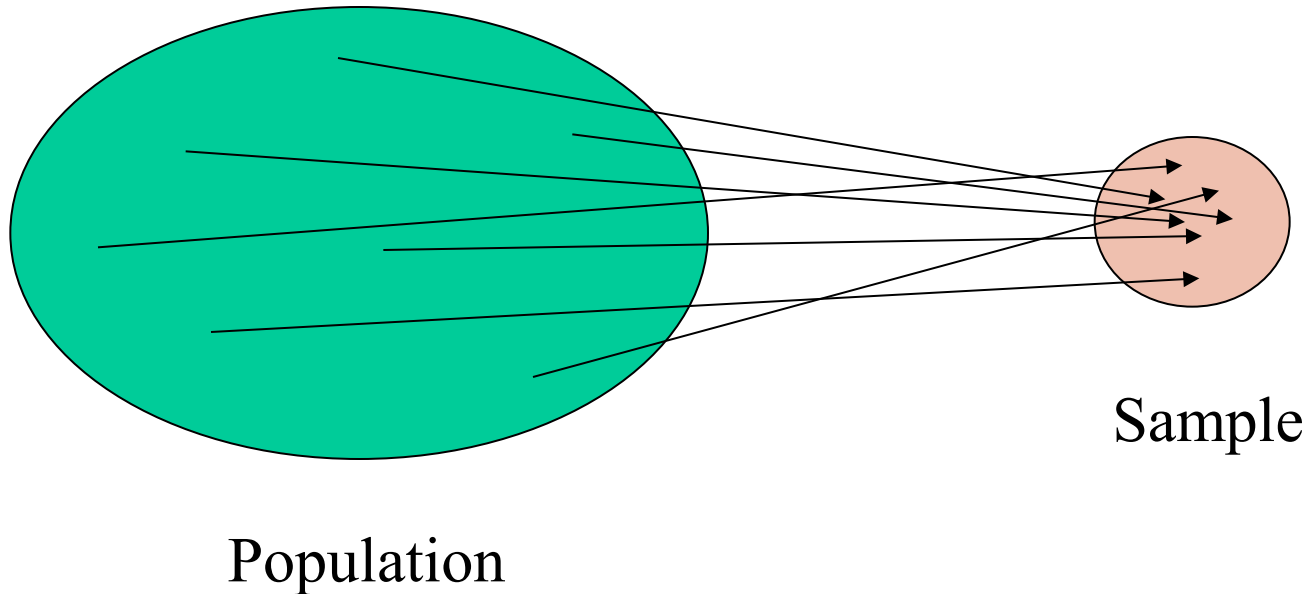
Question: Can aspirin reduce the risk of heart attack in humans?

- **Sample:** Sample of 22,071 male physicians between the ages of 40 and 84, randomly assigned to one of two groups. One group took an ordinary aspirin tablet every other day (headache or not). The other group took a placebo every other day. This group is the control group.
- **Summary statistic:** The rate of heart attacks in the group taking aspirin was only 55% of the rate of heart attacks in the placebo group.
- **Inference to population:** Taking aspirin causes lower rate of heart attacks in humans.

SAMPLING A SINGLE POPULATION

- **Sampling Techniques**

- **Simple Random Sample (SRS):** every member of the population has an equal chance of being selected.



- **Simple Random Sample**



IS IT THAT EASY?

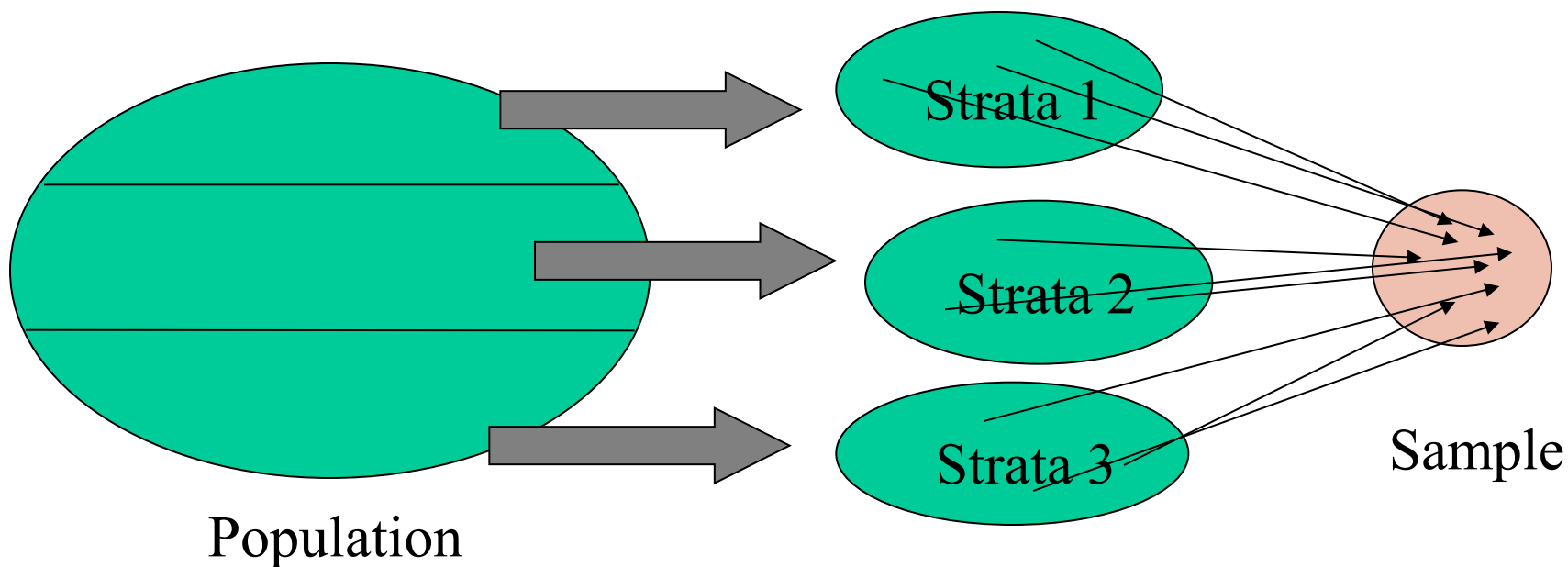


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SAMPLING A SINGLE POPULATION

- **Sampling Techniques**

- **Stratified Random Sample:** Divide the sample into several strata. Then take a SRS from each stratum.

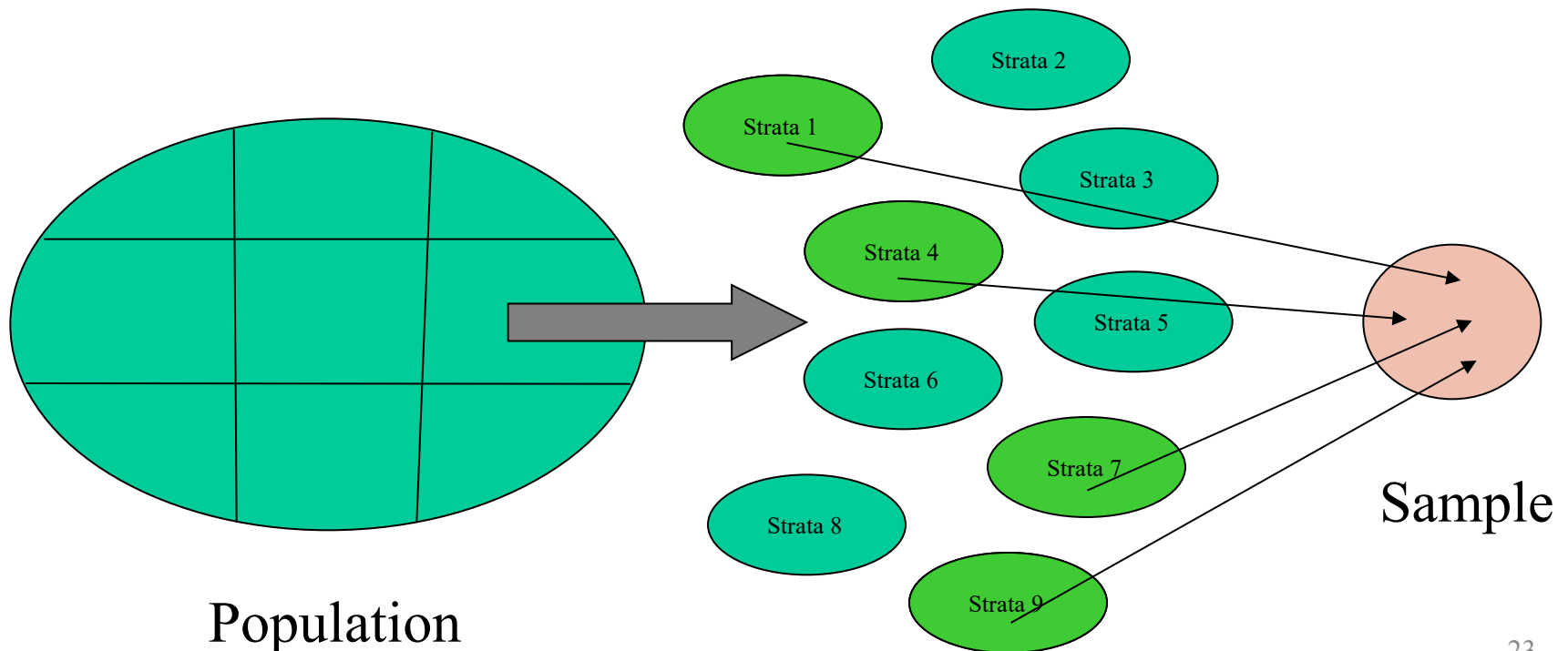


- **Advantage:** Each stratum is guaranteed to be randomly sampled
- **Example:** Obtain a list of all SSN for individuals in the U.S. who are over 65. Divide up the SSNs into region of the country (time zones). Then randomly sample 30 from each time zone.

SAMPLING A SINGLE POPULATION

- **Sampling Techniques**

- **Cluster Sample:** Divide the sample into several strata or clusters. Then take a SRS of clusters.



SAMPLING A SINGLE POPULATION

- **Sampling Techniques**

- **Cluster Sample**

- **Advantage:** May be the only feasible method, given resources.
 - **Example:** Obtain a list of all SSNs for individuals in the U.S. who are over 65. Sort the SSNs by the last 4 digits making each set of 100 a cluster. Use a random number table to pick the clusters. You may get the 4100's, 5600's and 8200's for example.



INFERENCE OVERVIEW

- **Describing a Population**

- It is common practice to use Greek letters when talking about a population.
- We call the mean of a population μ .
- We call the standard deviation of a population σ and the variance σ^2 .
- When we are talking about percentages, we call the population proportion π (or pi).
- It is important to know that for a given population there is only **one** true mean and **one** true standard deviation and variance or **one** true proportion.
- There is a special name for these values: **parameters**.



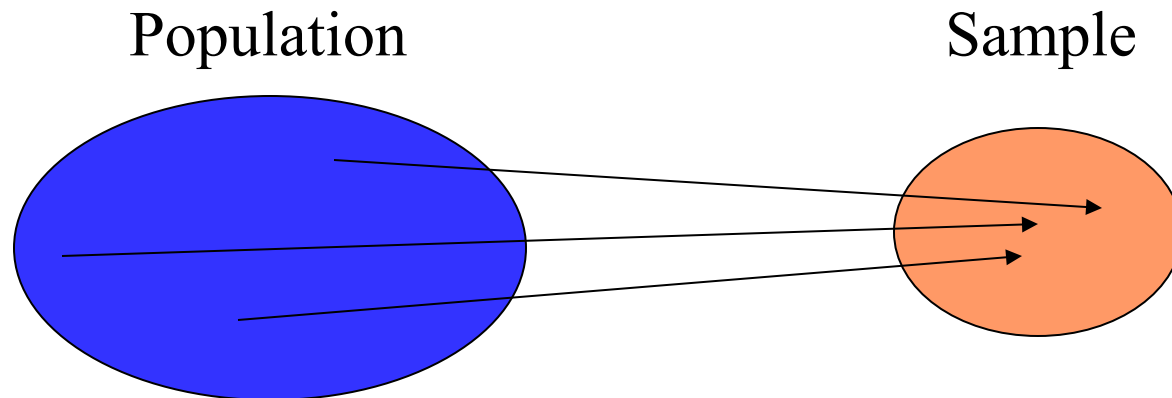
INFERENCE OVERVIEW

- **Describing a Sample**

- We call the mean of a sample $\bar{x}$.
- We call the standard deviation of a sample s and the variance s^2 .
- When we are talking about percentages, we call the sample proportion $\hat{\pi}$.
- There are many different possible samples that could be taken from a given population. For each sample there may be a **different** mean, standard deviation, variance, or proportion.
- There is a special name for these values: **statistics**.

INFERENCE OVERVIEW

- We use sample statistics to make inference about population parameters



Mean:	μ	$\bar{x}$
Standard Deviation:	σ	s
Proportion:	π	$\hat{\pi}$